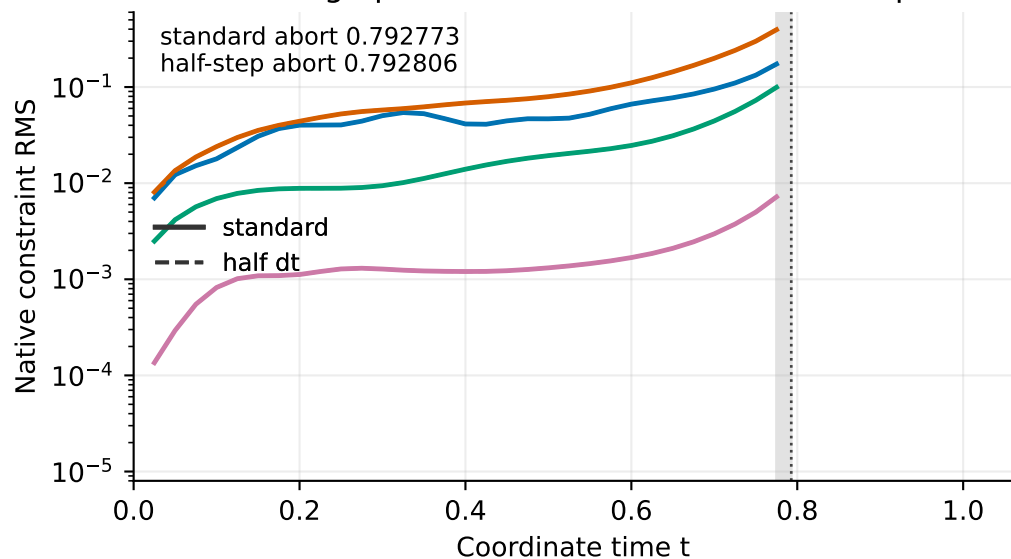


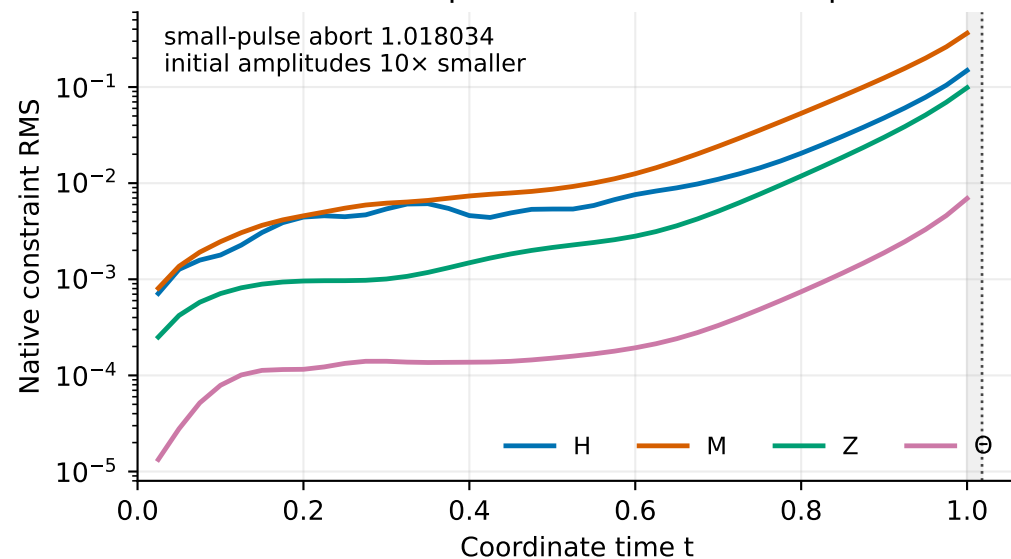
N24 native reference-wave-map controls — stopped-run observations

Same Cartesian spherical grid, wide reference (0.05–0.95), $a=0.5$, $\kappa=10$, symmetric degree-2 ghosts, $KO=0.1$

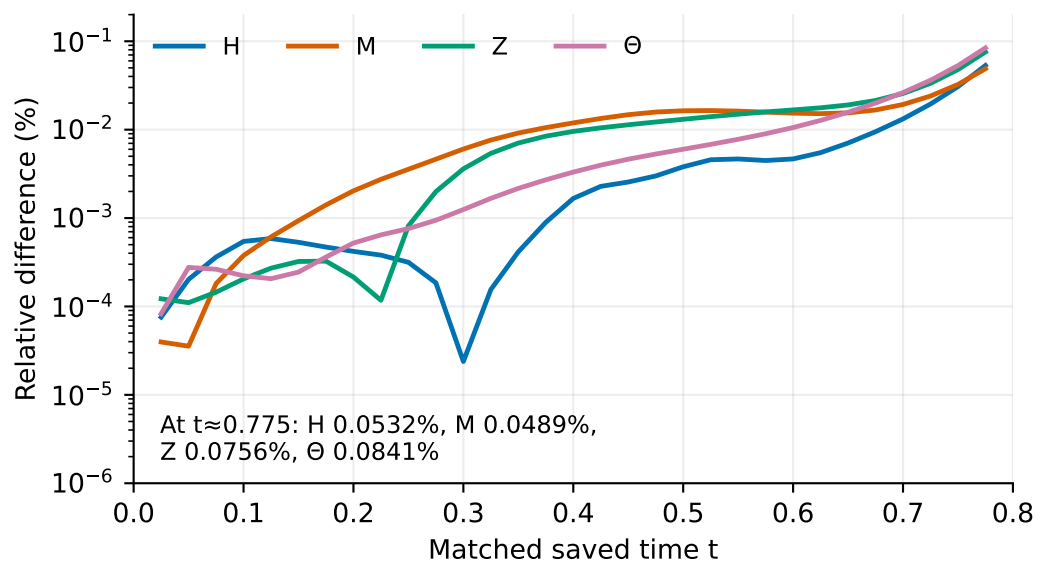
A Large pulse: standard and half timestep



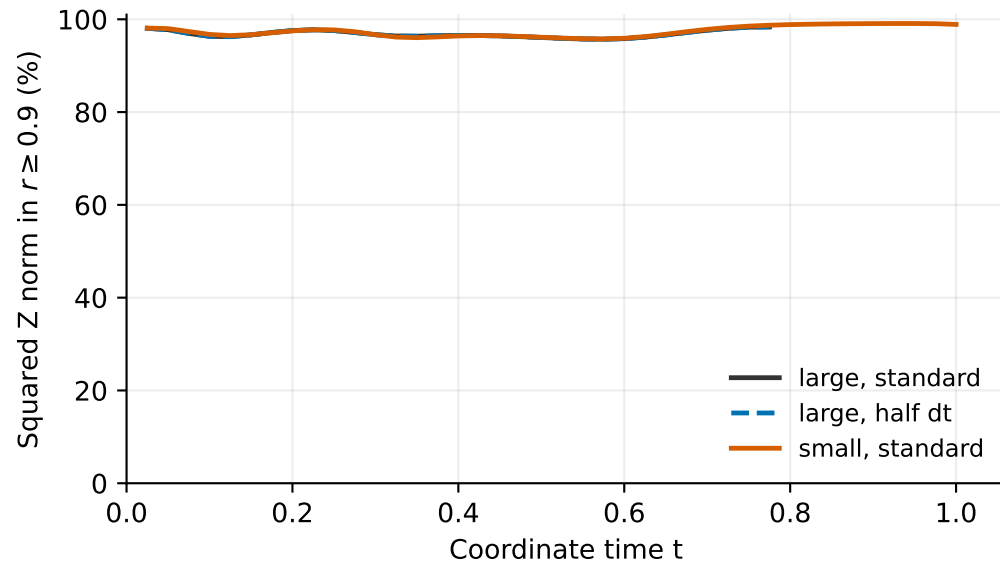
B Small pulse: standard timestep



C Half vs standard: saved RMS differences



D Saved Z localization



All three targeted $t=2$ and aborted at the same first reported cell ($\Omega=0.00217014$). Curves stop at saved times; shading marks the interval to the unsaved abort. All 105 saved states passed the recorded field guards.

Log panels omit $t=0$ near-roundoff norms. Shell fractions use unchanged native contractions and active counts.